

Research paper

Edge-enabled smart agriculture framework: Integrating IoT, lightweight deep learning, and agentic AI for context-aware farming

Muhammad Usman Tariq^a, Sheikh Muhammad Saqib^b, Tehseen Mazhar^{c,i,*},
Muhammad Amir Khan^d, Tariq Shahzad^{e,*}, Habib Hamam^{f,g,h,i}

^a Department of Marketing, Operations, and Information Systems, College of Business, Abu Dhabi University, Abu Dhabi, United Arab Emirates

^b Department of Computing and Information Technology, Gomal University, Dera Ismail Khan 29050, Pakistan

^c School of Computer Science, National College of Business Administration and Economics, Lahore 54000, Pakistan

^d School of Computing Sciences, Faculty of Computer Science and Mathematics, Universiti Teknologi Mara, Shah Alam, 40450 Selangor, Malaysia

^e Department of Computer Engineering, COMSATS University Islamabad, Sahiwal Campus, Sahiwal, 57000, Pakistan

^f Faculty of Engineering, Uni de Moncton, Moncton, NB E1A3E9, Canada

^g School of Electrical Engineering, University of Johannesburg, Johannesburg 2006, South Africa

^h International Institute of Technology and Management (IITG), Av. Grandes Ecoles, Libreville BP 1989, Gabon

ⁱ College of Computer Science and Engineering (Invited Prof.), University of Ha'il, Ha'il, SA 55476, Saudi Arabia

^j Department of Computer Science and Information Technology, School Education Department, Government of Punjab, Layyah 31200, Pakistan

ARTICLE INFO

Keywords:

Agentic AI
Edge computing
Internet of things (IoT)
Low-power deep learning
Precision farming
Smart agriculture
Weather and crop classification

ABSTRACT

Smart farming in connectivity-limited, energy-sensitive environments demands on-device perception and decision-making to reduce latency and cloud dependence. This article proposes an edge-enabled smart agriculture framework that integrates lightweight deep learning, rule-based agentic AI, and Internet of Things (IoT) devices for real-time, autonomous farming decisions. The system features two vision-based models—one for weather classification and one for crop identification—built on the MiT-B0 Vision Transformer architecture and optimized for low-resolution (128×128) image inputs. These models run on resource-constrained hardware suitable for rural deployment and support efficient, on-device processing. Weather prediction spans 11 classes (e.g., frost, lightning, rain, sandstorm), while crop classification covers 5 major crops. The system achieves an accuracy of 88 % for weather and 93 % for crops, with high F1-scores and low MAE, Kappa, and Hamming loss values.

Predictions are interpreted by a rule-based agentic AI layer that triggers actions across multiple IoT actuators, such as smart irrigation, NDVI sensors, frost alarms, drones, and pest detectors. The decision engine supports both joint rule logic (e.g., activating hail protectors when hail is detected in maize fields) and fallback single-condition rules. Python-implemented case studies show seamless model-AI-IoT interaction in combined and separate scenarios.

By minimizing cloud dependency, reducing communication overhead, and enabling low-power operation, the proposed framework addresses critical challenges in connectivity-limited, energy-sensitive agricultural contexts. It demonstrates the potential for scalable and intelligent smart farming, aligning with the goals of sustainable Agriculture 4.0.

1. Introduction

1.1. Context

In response to global pressures such as population growth, climate change, and the reduction of arable land, agriculture must rapidly

evolve toward intelligent and sustainable production systems. According to the United Nations Food and Agriculture Organization [1], the global population is projected to reach 10 billion by 2050, and could exceed 11 billion by the end of the century [2]. This demographic surge demands a 70 % increase in food production [3], yet the global area of cultivable land continues to shrink—from around 20 million to 18

* Corresponding authors.

E-mail addresses: tehsenmazhar719@gmail.com (T. Mazhar), tariq@cuisahiwal.edu.pk (T. Shahzad).

<https://doi.org/10.1016/j.rineng.2025.107342>

Received 11 August 2025; Received in revised form 6 September 2025; Accepted 16 September 2025

Available online 17 September 2025

2590-1230/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

million square miles—due to urban expansion and industrial development [4]. These opposing trends underscore the urgency of modernizing agriculture using digital and autonomous technologies.

Traditional agricultural methods—largely labor-intensive and reliant on experiential knowledge—struggle to adapt to climate volatility, resource constraints, and scalability requirements. Challenges such as unpredictable weather, erratic soil health, and inefficient input usage (e.g., water, fertilizers, pesticides) further compound the sector's environmental and economic burdens [5,6]. While past agricultural phases—from manual operations (Agriculture 1.0) to mechanization and basic automation (Agriculture 3.0)—have improved productivity, they lack the intelligent, real-time decision-making capabilities that define the emerging paradigm of Agriculture 4.0.

To operationalize the goals of Agriculture 4.0 under intermittent connectivity and tight energy/bandwidth budgets, we adopt an edge-first architecture centered on two compact vision tasks—weather and crop identification—executing directly on resource-constrained devices. This dual-vision design is preferred over heavier multi-sensor fusion because it provides the minimal, actionable context needed for farm-level decisions while keeping hardware, integration, and maintenance overheads low. In particular: (i) Context sufficiency and actionability—crossing a predicted weather state with the detected crop type maps directly to interventions (e.g., frost alert for wheat, hail protection for maize) without requiring external data streams; (ii) Edge suitability—low-resolution inputs (128×128) and a lightweight MiT-B0 transformer backbone enable low-latency, low-power inference with a global receptive field that preserves scene-level cues important for both sky conditions and field structure; (iii) Robustness and maintainability—avoiding heterogeneous sensor fusion removes calibration/synchronization failure modes and allows each vision model to be updated, quantized, or replaced independently; (iv) Cost and privacy—commodity cameras and on-device processing reduce bill of materials, bandwidth usage, and exposure of farm data. This rationale directly motivates the proposed system layout, in which the dual-vision outputs feed a rule-based agentic layer that triggers IoT actuators for closed-loop, context-aware control at the edge.

Despite increasing investments in rural Internet connectivity, many farming regions still face poor infrastructure, intermittent coverage, and energy limitations. To meet these challenges, this study proposes an edge-deployable, autonomous smart agriculture framework that fuses vision-based deep learning with agentic artificial intelligence and a network of real-time Internet of Things (IoT) devices. The system is built around two lightweight, transformer-based models: one trained for crop classification and another for weather condition detection. Both models use the efficient NVIDIA/mit-b0 backbone and operate on reduced-resolution (128×128) input images, enabling deployment in resource-constrained, low-power environments.

Live imagery from field-deployed cameras is processed every hour, generating real-time predictions. These are interpreted by a rule-based agentic AI layer that translates environmental and agronomic insights into automated actions via connected IoT actuators—such as irrigation controllers, drones, pest detectors, and environmental sensors. This closed-loop architecture transforms passive sensing into proactive field management, minimizing manual labor, reducing input waste, and increasing resilience to climate variability. The proposed system offers a scalable, cost-effective, and intelligent solution aligned with the goals of Agriculture 4.0 and the growing demands of sustainable food production.

1.2. Gap analysis: why existing edge-AI solutions fall short on contextual fusion

Despite notable progress, three persistent limitations motivate our design. First, context is rarely fused on-device: many edge deployments treat perception as single-task (e.g., crop or disease only) and defer cross-context reasoning—such as combining field state with imminent

weather—to cloud services or manual post-processing, which is fragile under rural connectivity constraints. Second, when contextual fusion is attempted, it often relies on heterogeneous sensors whose calibration, time-synchronization, and bandwidth needs are brittle and costly to maintain at farm scale. Third, closed-loop autonomy is seldom realized: systems frequently stop at reporting classifier metrics without a verified, auditable path to actuation.

Our approach addresses these gaps as follows: (i) Context sufficiency at low cost—crossing two camera-derived signals (weather class and crop identity) yields actionable context for interventions (e.g., frost alert for a detected crop) without adding specialized sensors; (ii) Edge feasibility—compact MiT-B0 models operating on 128×128 inputs execute efficiently on resource-constrained hardware, reducing latency and bandwidth and avoiding cloud dependence; (iii) Robustness and maintainability—two decoupled vision models eliminate multi-sensor calibration overhead and allow independent updates, quantization, or replacement; (iv) From perception to action—a rule-based agentic layer converts joint and fallback conditions into IoT actuation (e.g., irrigation control, frost alarms, drone dispatch), validated through Python-implemented case studies; (v) Privacy and scalability—on-device inference limits data exposure while enabling replication across fields without network upgrades.

1.3. Objectives

The primary objectives of this research are to design and implement a real-time, intelligent smart agriculture framework that meets the challenges of limited connectivity, energy constraints, and operational complexity in rural and resource-constrained farming environments. Specifically, the system aims to:

- Develop an autonomous, edge-deployable smart agriculture system that seamlessly integrates deep learning, agentic AI, and Internet of Things (IoT) infrastructure for in-field decision-making.
- Classify crop types and assess weather conditions using two lightweight, independently trained models based on the MiT-B0 Vision Transformer backbone optimized for low-resolution inputs.
- Implement a rule-based agentic AI engine that interprets model outputs to trigger precise, context-aware actions via a suite of connected IoT devices.
- Ensure low-latency, energy-efficient inference and system scalability to support real-time actuation in connectivity-limited or serverless deployment scenarios.

1.4. Key contributions

The main contributions of this work are summarized as follows:

- An edge-first smart agriculture framework that unites dual vision models (weather and crop) with a rule-based decision layer and IoT actuation for real-time, on-device control in connectivity- and energy-constrained settings.
- Lightweight MiT-B0 backbones at 128×128 enable CPU-only inference for 11 weather classes and 5 crop classes, supporting low-latency operation without cloud dependence.
- Demonstrated performance: crop model accuracy 93 % ($F1 = 0.93$; $MAE = 0.1116$) and weather model accuracy 88 % ($F1 = 0.88$; $MAE = 0.5277$), confirming reliable edge inference.
- A rule-based agentic AI layer that fuses joint (weather $\times$ crop) and fallback (single-input) logic to trigger IoT devices (e.g., irrigation controllers, NDVI sensors, pest traps, frost alarms) with auditable trace logs and safety guardrails.
- A modular, interpretable, and deployment-ready pipeline validated via Python case studies on a CPU-only workstation, minimizing bandwidth usage and simplifying maintenance.

- A clear gap-bridging rationale: on-device contextual fusion and closed-loop autonomy at the edge to address limitations of prior detection-only or cloud-dependent designs.

1.5. Composition of the article

The remainder of this article is organized as follows. [Section 2](#) presents the literature review, outlining recent developments in smart agriculture and identifying key gaps that motivate this research. [Section 3](#) describes the proposed methodology, including the architecture and training of the weather and crop classification models, the agentic AI decision-making mechanism, and the integration with IoT devices. [Section 4](#) provides the results and discussion, including experimental validation, statistical performance, applied examples, and comparisons with related works. Finally, [Section 5](#) concludes the study and discusses future research directions to enhance the scalability and adaptability of the proposed framework.

2. Literature review

2.1. Current research

Despite notable advancements in agriculture, prior efforts have shown several limitations when compared to the capabilities of the proposed system. Initial strategies focused on enhancing yield through the extensive use of seeds, fertilizers, and pesticides [2]. However, these traditional methods have proven inadequate in meeting today's food demand. To address this, genetically modified foods were introduced [4], but research has highlighted serious health concerns, including infertility, premature aging, and weakened immune systems [7], which led to public skepticism and a preference for natural, organic alternatives. In recent years, smart agriculture techniques—such as vertical farming, hydroponics, and aeroponics—have emerged, integrated with IoT technologies to improve crop yield and quality [8]. While these approaches have seen some success, they often require high infrastructural investment and lack intelligent, autonomous decision-making systems that adapt to real-time environmental changes. Additionally, existing models are not always scalable or practical for deployment in resource-limited settings like rural or underdeveloped regions [9,10].

For example, Khan et al. [11] examined the integration of drone-based imaging and real-time IoT monitoring for sustainable farming and identified critical barriers related to data management, cost, and regulatory frameworks. Their findings highlight the importance of intelligent systems that can offer automated, context-aware decisions in agricultural settings. Similarly, Selvanarayanan et al. [12] developed an RNN-driven IoT system for soil quality optimization in coffee farming, leveraging counterfactual AI recommendations for irrigation and fertilization, which aligns with the objectives of enhancing decision support in precision agriculture.

Despite extensive research and technological development in IoT-based nutrient and crop management systems, most existing approaches remain reactive and narrowly focused. Prior studies have successfully introduced precision farming innovations using automated devices [13], drones [13], satellite imagery [14], big data analytics [15], and machine learning algorithms [16] to improve agricultural productivity and sustainability. For instance, Hartono et al. developed a portable IoT system to monitor soil nutrients like nitrogen, phosphorus, and potassium, enabling farmers to make informed fertilizer decisions [17]. Similarly, Islam et al. (2023) presented an advanced machine learning-based IoT device that provides real-time soil monitoring and crop-specific recommendations, thereby optimizing input usage and yield outcomes [18]. These advancements have enabled localized precision but still lack the ability to make context-aware decisions autonomously.

Weather forecasting has also been enhanced by AI-driven approaches. Kaushik et al. [19] proposed an artificial neural

network-based weather prediction model, achieving 90.4 % accuracy using smart computing devices. Their work underscores the potential of neural models in delivering efficient predictions in resource-constrained environments, which supports our choice of a lightweight ViT-based weatherModel.

Recent research in smart agriculture has explored diverse AI and IoT-driven approaches to crop forecasting, disease detection, and multi-modal data processing. However, a significant limitation across the latest studies is the absence of a unified, real-time, lightweight, and decision-driven framework that fuses crop and weather imagery at the edge. For instance, [20] provides a comprehensive review of CNN-based agricultural applications but does not demonstrate an end-to-end, real-time system integrating crop and weather detection. Similarly, [21] emphasizes irrigation and disease monitoring techniques yet overlooks continuous fusion of crop-weather context and agentic automation in IoT systems. To address this critical gap, the proposed model introduces an Agentic AI-driven IoT framework that combines dual visual inference models (for crop type and weather condition) with a rule-based decision engine, enabling real-time, on-field actuation.

In [22], UAV-based solutions for anomaly detection in crops rely heavily on remote sensing and cloud computation, making them unsuitable for edge deployments. The proposed work diverges by using fixed, low-cost cameras and 128×128 input resolution to facilitate on-device computation. While [23] attempts knowledge distillation for on-device systems, it lacks integration of weather context and a decision engine, which the proposed model provides using cropModel and weatherModel as synchronized inputs. In contrast to [24], which relies on large language models (LLMs) for reasoning, the proposed system avoids high resource consumption by leveraging an efficient nvidia/mit b0 backbone and a rule-based actuator, reducing infrastructure requirements.

Furthermore, blockchain-based forecasting in [25] addresses data security but omits real-time decision-making from visual inputs. In contrast, Ktari et al. [26] proposed a multi-sensor traceability system for olive oil using embedded IoT platforms and lightweight blockchain smart contracts, illustrating the viability of decentralized solutions in agricultural monitoring. While Our Article does not focus on blockchain, the study emphasizes low-power, sensor-integrated agricultural frameworks aligned with our model's lightweight deployment goals. Research such as [27] focuses narrowly on rice disease detection using GAN/RNN architectures without supporting multicrop or multimodal conditions. The proposed framework is scalable across crop types and integrates a rule engine for decision-making.

In [28], the focus remains on generalized AI-IoT synergy without implementing a visual fusion pipeline. The proposed work stands out by establishing a transparent pipeline—from dual-model inference to real-time action. Sensor-based yield prediction approaches like [29] do not utilize image-based crop-weather detection or support autonomous control. The proposed model operates directly on visual cues for timely IoT responses.

Although [30] discusses computer vision in agriculture, it lacks deployment of an integrated crop-weather control model. Lastly, [31] presents generic challenges such as cost and infrastructure without offering concrete, deployable solutions. The proposed system addresses these concerns through lightweight deployment, efficient edge operation, and a practical, interpretable framework to enhance decision-making in smart agriculture environments.

To address these critical gaps, the proposed research introduces an Agentic AI-driven IoT framework for agriculture that integrates crop detection and weather forecasting to enable real-time, autonomous decision-making in smart farming environments.

The proposed system fulfills the limitations observed in prior research by combining deep learning, agentic intelligence, and IoT technologies into a low-cost, interpretable, and edge-deployable framework. It supports real-time crop health assessment and environment-aware actuation by using dual visual models for crops and

weather, a lightweight decision engine, and field-level automation. This comprehensive design ensures the proposed model directly addresses the research gaps discussed above—bridging the divide between detection, contextual awareness, and timely, intelligent responses in agriculture.

2.2. Research gap

While prior studies have introduced AI and IoT-driven approaches for agriculture, several critical limitations persist that hinder their real-world applicability and integration. Most existing frameworks focus on either crop or weather data in isolation and lack a unified architecture capable of making real-time, context-aware decisions based on fused multimodal inputs. This siloed approach restricts the ability to optimize agricultural actions that require concurrent environmental and crop-specific insights.

Additionally, previous works often rely on cloud-based models or high-complexity architectures such as CNN-LSTM hybrids or transformer-based large language models, which are unsuitable for edge deployment in rural or resource-constrained environments. These solutions fail to meet the real-time inference needs of smart agriculture scenarios that demand lightweight, energy-efficient systems operating at the edge.

Another critical gap lies in the limited use of agentic AI for autonomous decision-making. Most IoT-based agriculture systems remain reactive, requiring manual intervention or rule tuning. They do not leverage rule-based automation that interprets deep learning model outputs to trigger specific device actions. This absence of intelligent, proactive actuation reduces system efficiency and scalability.

Moreover, while some blockchain-based or federated learning frameworks offer data traceability and privacy, they do not integrate visual perception models for real-time agricultural state assessment. Consequently, they miss the opportunity to connect real-time crop and weather detection with IoT-enabled actuation.

In response to these gaps, the proposed research introduces a lightweight, unified framework that integrates dual deep learning models for crop and weather classification, a rule-based agentic AI decision engine, and an IoT actuation layer. This system uniquely combines visual intelligence with real-time autonomous control, optimized for edge deployment in precision farming applications.

3. Proposed methodology

The development of the proposed smart agricultural system is structured around a two-step methodology, aimed at delivering intelligent, sustainable, and scalable solutions for modern agriculture. The initial step encompasses the meticulous training of two specialized deep learning models using dedicated datasets. Following this, the second step leverages the predictions generated by these trained models to simulate agentic AI decisions, which subsequently control smart agricultural IoT devices for automated actions.

3.1. Agentic AI decision layer

In this work, we use “agentic” in its minimal, operational sense: an autonomous loop that (1) perceives state from local models and sensors, (2) maps that state to actions under explicit policies, and (3) actuates devices in the environment—then re-evaluates upon new inputs. Concretely, our layer ingests the dual-vision outputs (weather class $\times$ crop identity) and converts them into actuator commands (e.g., irrigation control, frost alarm, drone dispatch) using deterministic policies.

What makes it agentic rather than a static expert system is the closed perception $\rightarrow$ decision $\rightarrow$ actuation loop running on-device: actions are executed (not merely recommended), and the policy is applied continuously as new frames/events arrive, enabling state-dependent behaviors (e.g., joint conditions vs. fallback single-condition rules) at the edge.

We do not claim learning-to-plan or stochastic policy optimization in this layer. In our implementation, “agentic” refers to autonomy of action under explicit, auditable policies deployed at the edge. This aligns with the system’s goal of reliable, low-latency, connectivity-independent control while preserving maintainability and traceability.

3.2. Hardware and runtime environment for case studies

All Python case studies were executed on Google Colaboratory with the following configuration: CPU: Intel Xeon-class virtual processors; System memory: 12–16 GB; Operating system: Ubuntu 20.04 LTS (Colab backend); Python: 3.10. Key libraries: PyTorch 2.1.0 with torchvision 0.15.x, transformers 4.30.x, scikit-learn 1.3.x, NumPy 1.24.x, and Matplotlib 3.7.x. Hardware acceleration: NVIDIA Tesla T4 GPU with CUDA/cuDNN enabled. The crop and weather classification models, together with the rule-based decision logic, were implemented and executed in Google Colaboratory, ensuring that all hardware and software requirements for training and inference were satisfied without the need for additional local resources.

3.3. Edge-deployment prototype protocol (Raspberry Pi / Jetson Nano)

This protocol describes how to run the trained MiT-B0 weather and crop models and the rule-based decision layer on Raspberry Pi 4 (64-bit OS) and NVIDIA Jetson Nano (JetPack). It does not introduce hardware performance measurements.

(1) Export both trained models to ONNX with static input $1 \times 3 \times 128 \times 128$; keep FP32 as the reference; optional post-training quantization may be applied with the target runtime’s tooling. (2) On Raspberry Pi, install ONNX Runtime (CPU execution provider) plus NumPy and OpenCV; on Jetson Nano, build a TensorRT engine from the ONNX export (or use ONNX Runtime) with identical pre/post-processing. (3) Use a single Python entry point that loads both ONNX models, applies $\text{resize} \rightarrow \text{center-crop} \rightarrow \text{normalize}$ to 128×128 , runs batch-1 inference, and forwards top-1 labels with confidences to the policy engine. (4) Drive actuators via GPIO libraries (gpiozero/RPi.GPIO on Raspberry Pi; Jetson.GPIO on Jetson), mapping policy actions to relays/alarms. (5) Maintain configuration parity with the workstation case studies: identical class lists, thresholds, and the YAML/JSON policy table. (6) Validate functionality by replaying a small held-out sample to check label/confidence consistency, correct rule selection (joint vs. fallback), emitted commands, and trace logs (timestamp, labels, confidences, rule ID, command). (7) Operate offline by default; if optional telemetry is enabled, restrict it to derived signals over authenticated, encrypted channels; store logs locally with configurable retention. (8) Apply change control to policy updates: $\text{lint} \rightarrow \text{validate} \rightarrow \text{re-run functional tests} \rightarrow \text{deploy}$; if a policy file is missing or invalid, fall back to monitor/alert-only mode..

3.4. Rule elicitation and policy encoding

We derive executable policies that map dual-vision outputs (weather class, crop identity) to IoT actions under explicit priorities (safety $\rightarrow$ yield $\rightarrow$ resource conservation):

Step 1 — Inventory and action primitives. We enumerate weather classes (11), crop classes (5), available actuators (irrigation, NDVI probe readout, frost alarm, drone dispatch, pest detector), and define atomic actions (on/off, set-point adjustment, scheduled check, alert).

Step 2 — Risk-action matrix. We construct a decision table crossing weather $\times$ crop to an action intent with a default non-intervention row; entries specify action type and priority level. Ambiguous cells resolve to alerts or monitoring-only.

Step 3 — Trigger specification and guardrails. For each entry we specify triggers on model outputs (top-1 label and confidence), joint rules when both models are confident, fallback rules when one signal is missing/low-confidence, tie-breakers by priority, and guardrails to

prevent oscillation (debounce window, minimum on/off duration, and rate limiting).

Step 4 — Table-driven verification. We validate rules using Python case studies that replay canonical scenarios and adversarial edge cases (conflicting triggers, rapid weather shifts, low-confidence frames). Tests assert expected actuator states and produce trace logs for auditability.

Step 5 — Policy encoding and deployment. The decision table is stored in a human-readable YAML/JSON schema and loaded at startup; the engine evaluates rules in priority order and emits commands to device drivers. Unit tests cover parsing, evaluation order, and safety interlocks.

Step 6 — Change management. Rule updates follow a minimal change-control procedure (edit → lint/validate → re-run tests → deploy), with version tags recorded alongside results.

3.5. Backbone selection and baseline scope

We adopt a single lightweight transformer backbone (MiT-B0) for both weather and crop classifiers to standardize pre-processing (128×128), quantization options, and CPU-only deployment at the edge. This choice reduces integration and maintenance overhead by sharing infrastructure across tasks while retaining a global receptive field useful for scene-level cues. Although widely used lightweight CNN backbones (e.g., MobileNet-style and EfficientNet-Lite-style) are relevant for edge inference, we did not benchmark them in this study to preserve a focused scope and to avoid changing the reported training/evaluation budget. Importantly, the pipeline is backbone-agnostic: heads and loaders expose a uniform interface, enabling drop-in backbones to be trained/evaluated under identical input resolution, augmentation, and optimization settings for fair, like-for-like comparisons in future extensions.

3.6. Preparing well-trained deep learning models

This foundational phase is critical for establishing the core intelligence of the entire system. It involves the preparation and subsequent training of two lightweight and highly efficient deep learning models: a weatherModel for predicting weather conditions and a cropModel for classifying crop health. Both models are built upon the nvidia/mit-b0 backbone, a choice optimized for resource-constrained environments to ensure rapid deployment and inference capabilities in real-world field conditions. The comprehensive process for preparing these models is visually represented in Fig. 1.

As illustrated in Fig. 1, the process commences with the acquisition of distinct Weather Image Datasets and Crop Image Datasets. These datasets undergo specific image transformations, including resizing to

128×128 pixels and normalization utilizing precisely tuned parameters (mean = [0.48,0.47,0.42], std = [0.226,0.225,0.227]). The training regimen employs a learning rate of 1×10^{-4} , a momentum of 0.9, and a weight decay of 1×10^{-5} . A batch size of 32 is consistently applied, with 4 workers facilitating efficient data loading. The Adam optimizer is selected for model optimization. Each model is trained for 10 epochs, culminating in the saving of the resulting trained models as weatherModel.pth and cropModel.pth, respectively.

3.6.1. Training weather deep learning model on weather image dataset

The weather deep learning model is specifically trained to accurately classify various weather conditions based on input image data [32]. The training dataset designated for the weather model comprises a total of 3428 images, collectively representing 11 distinct weather types.

For consistency, we adopt standardized names in the text while noting the dataset's original labels on first mention (e.g., “fog/smog (dataset: ‘fogsmog’)”, “ground ice (dataset: ‘glaze’)”).

The comprehensive set of predicted classes for the weather model includes: dew, fog/smog (dataset: “fogsmog”), frost, ground ice (dataset: “glaze”), hail, lightning, rain, rainbow, rime, sandstorm, and snow.

The detailed distribution of training images across these individual classes indicates that the dataset includes 349 images for dew, 425 for fog/smog (dataset: “fogsmog”), 237 for frost, 319 for ground ice (dataset: “glaze”), 295 for hail, 188 for lightning, 263 for rain, 116 for rainbow, 580 for rime, 346 for sandstorm, and 310 for snow.

The training progression of the weather model is meticulously monitored through an analysis of loss and accuracy metrics across successive epochs. Fig. 2 graphically illustrates the reduction in training loss over time, providing clear evidence of the model's effective learning capabilities.

As depicted in Fig. 2, the training loss exhibits a rapid decrease during the initial epochs, subsequently stabilizing at a low value. This trajectory unequivocally demonstrates the model's proficiency in minimizing prediction errors. Concurrently, Fig. 3 showcases the consistent increase in training accuracy across the epochs, highlighting the model's improving performance in accurately classifying diverse weather conditions.

Fig. 3 distinctly indicates that the training accuracy steadily rises, ultimately achieving a high level. This robust performance signifies the model's strong learning and generalization capabilities when applied to the weather dataset.

3.6.2. Training crop deep learning model on crop image dataset

The crop deep learning model is specifically developed to assess crop health by accurately classifying different crop types from image-based

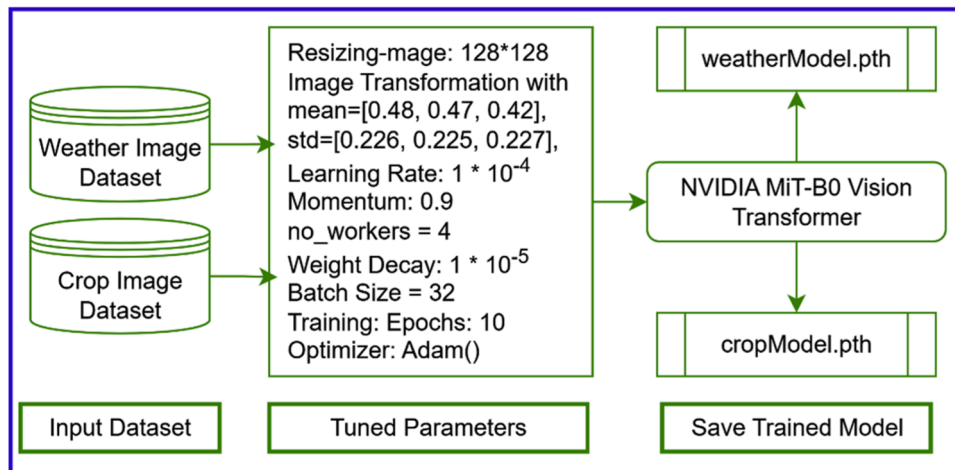


Fig. 1. Preparing deep learning trained models for weather and crop prediction.

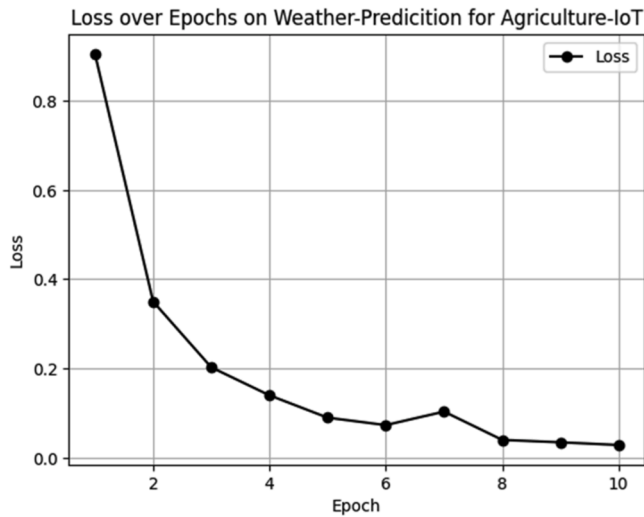


Fig. 2. Loss over epochs on weather-prediction for agriculture-IoT.

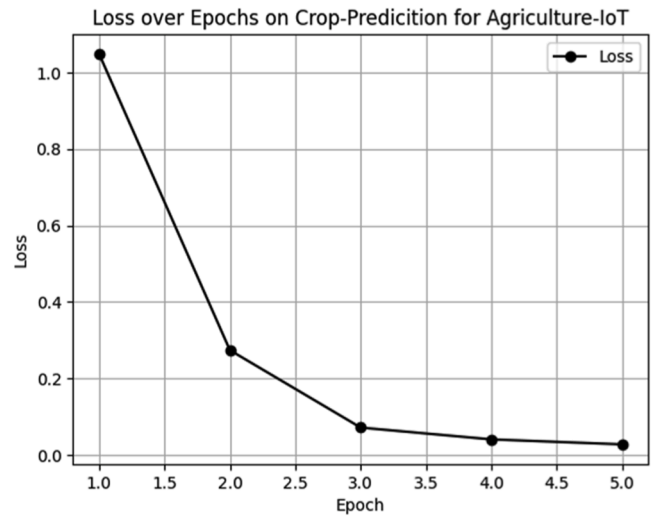


Fig. 4. Loss over epochs on crop-prediction for agriculture-IoT.

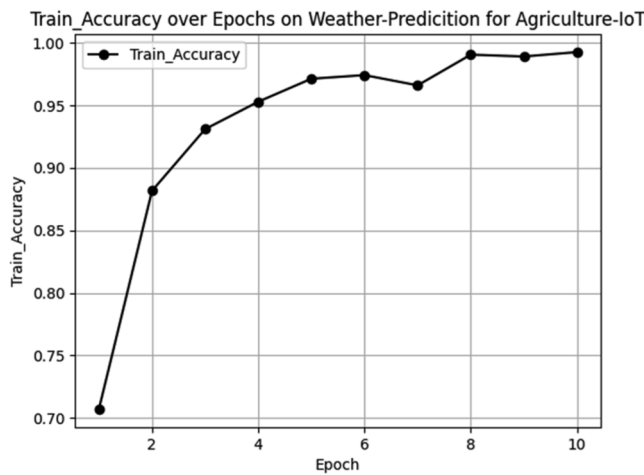


Fig. 3. Train_accuracy over epochs on weather-prediction for agriculture-IoT.

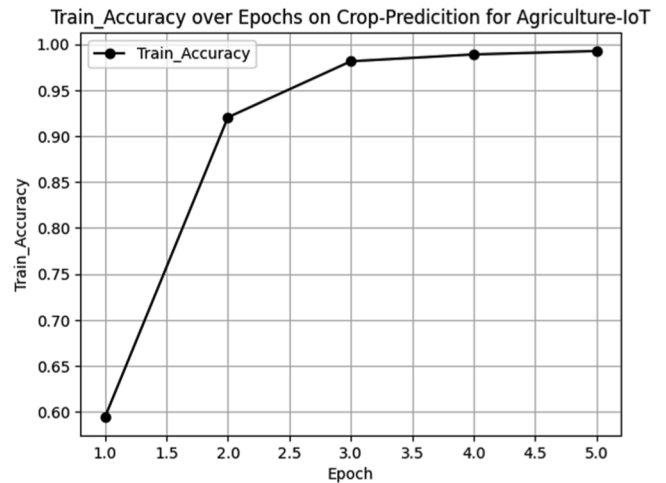


Fig. 5. Train_accuracy over epochs on crop-prediction for agriculture-IoT.

data [33]. The training dataset allocated for the crop model consists of a total of 804 images.

The distribution of training images across the various crop types shows that there are 160 images each for 'jute', 'maize', 'rice', and 'sugarcane', and 164 images for 'wheat'.

Similar to the weather model, the training performance of the crop model is rigorously evaluated by observing the trends in loss and accuracy over epochs. Fig. 4 visually represents the consistent reduction in loss during the training phase of the crop model.

Fig. 4 clearly demonstrates a significant decline in loss, particularly prominent in the early epoch, followed by stabilization at a minimal value. This pattern confirms the model's efficient learning process from the provided crop image dataset. Furthermore, Fig. 5 illustrates the notable improvement in training accuracy across the epochs for the crop model.

As depicted in Fig. 5, the training accuracy for the crop model exhibits a rapid increase, ultimately achieving a high level of accuracy. This performance underscores the model's strong capability in correctly identifying various crop types.

3.6.3. Validation of weather and crop deep learning models

To rigorously evaluate the efficacy and robustness of the trained weatherModel and cropModel, a comprehensive validation process was undertaken using dedicated test datasets. This section presents the

quantitative and qualitative results, demonstrating the superior performance of the proposed models in their respective classification tasks.

The test dataset for the Weather Model comprised a total of 3434 images. The distribution of these test images across the 11 weather classes was as follows: 349 for 'dew', 426 for 'fogsmog', 238 for 'frost', 320 for 'glaze', 296 for 'hail', 189 for 'lightning', 263 for 'rain', 116 for 'rainbow', 580 for 'rime', 346 for 'sandstorm', and 311 for 'snow'.

The test dataset of Crop Deep Learning Model having 251 images. The test set in this included 50 images of jute, 50 images of maize, 50 images of rice, 50 images of sugarcane and 51 images of wheat. These well balanced testing sets were carefully created to guarantee a fair and thorough testing and eventually exhibited the recommendable performance of the proposed models on all the classes that were to be tested.

3.6.3.1. Confusion matrix analysis. Initially, the performance of both deep learning models was evaluated with the help of confusion matrices that provide a detailed report on false and true classification pertaining to every class.

The results of Table 1 give the confusion matrix results of the weather deep learning model on the test data, presenting the most important metrics in each weather category, such as Precision, Recall, F1-Score, and Support, as well as the average of all of them, and their accuracy.

Table 1
Confusion matrix results for weather deep learning model on test data.

Weather Class	Precision	Recall	F1-Score	Support
Dew	0.95	0.92	0.93	349
Fog/Smog	0.94	0.90	0.92	426
Frost	0.67	0.82	0.74	238
Glaze	0.76	0.75	0.76	320
Hail	0.96	0.86	0.91	296
Lightning	0.98	1.00	0.99	189
Rain	0.91	0.85	0.88	263
Rainbow	0.97	0.97	0.97	116
Rime	0.84	0.91	0.88	580
Sandstorm	0.93	0.94	0.94	346
Snow	0.79	0.74	0.76	311
Average	0.88	0.88	0.88	3434
Accuracy	0.88			

The overall accuracy of the Weather Model reached the height of 0.88, and the average Precision, Recall and F1-Score also took on the value of 0.88. It was observed that other types like the Lightning and the Rainbow classes, performed enormously well with F1-Scores of 0.99 and 0.97 respectively, which signifies strong classification of such important weather conditions. Individually, although the F1-Scores of 'Frost', 'Glaze', and 'Snow' were slightly lower, they still can be applied to the real world setting, knowing that these are inferences of complexities of differentiating common real-worlds with images. Fig. 6 represents the confusion matrix of Weather Deep Learning Model on a visualized heatmap, which shows the confusion matrix clearly due to the true positives, false positives and false negatives.

Similarly, Table 2 presents the confusion matrix results for the Crop Deep Learning Model on its test data.

The Crop Model achieved an even higher overall accuracy of 0.93, with average Precision, Recall, and F1-Score of 0.94, 0.93, and 0.93 respectively. This indicates excellent performance across all crop types. Specifically, 'Wheat' demonstrated outstanding classification with an F1-Score of 0.97, and 'Jute' and 'Maize' also performed exceptionally well with F1-Scores of 0.95. The high recall for 'Rice' (0.98) is also notable, suggesting the model is highly effective at identifying this crop. Fig. 7 visually depicts the confusion matrix for the Crop Deep Learning Model, further illustrating its strong performance.

3.6.3.2. True and false prediction analysis. To further understand the

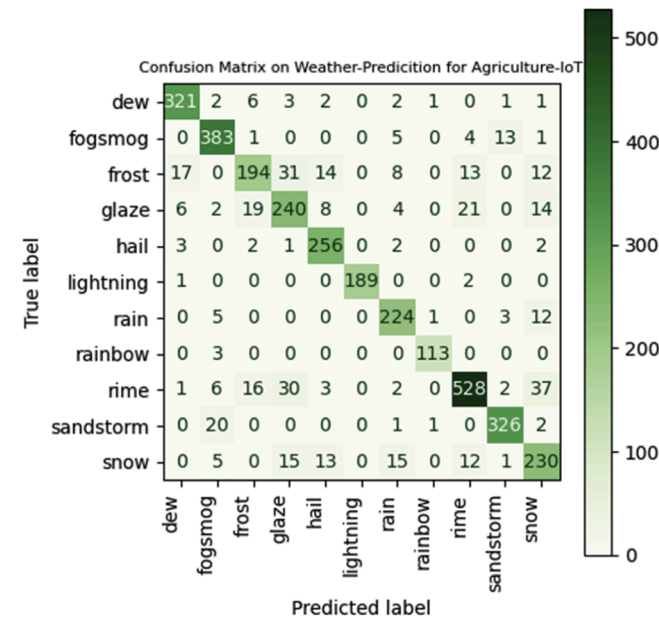


Fig. 6. Heat map for weather deep learning model.

Table 2
Confusion matrix results for crop deep learning model on test data.

Class	Precision	Recall	F1-Score	Support
Jute	0.94	0.96	0.95	50
Maize	1.00	0.90	0.95	50
Rice	0.79	0.98	0.88	50
Sugarcane	0.98	0.84	0.90	50
Wheat	0.98	0.96	0.97	51
Average	0.94	0.93	0.93	251
Accuracy	0.93			

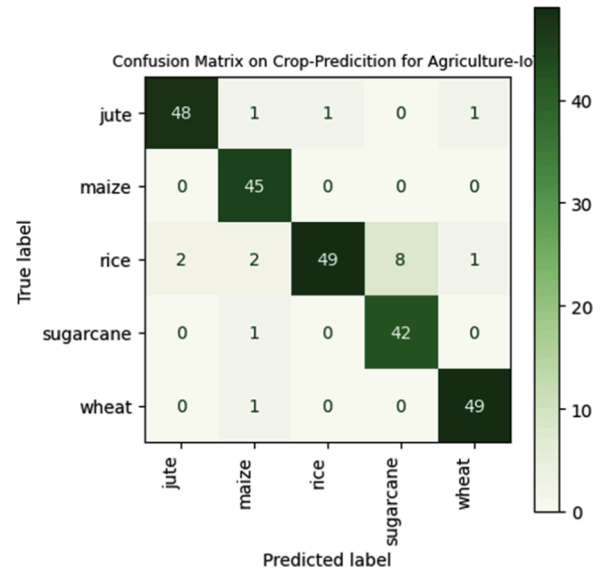


Fig. 7. Heat map for crop deep learning model.

models' predictive capabilities, the counts of true and false predictions for each class were analyzed. Fig. 8 presents the true and false predictions for the Weather Deep Learning Model.

As observed in Fig. 8, the number of true predictions (black bars) significantly outweighs the false predictions (red bars) across most weather categories, reinforcing the high accuracy reported in Table 1. This visual representation clearly indicates the model's strong ability to correctly identify various weather conditions.

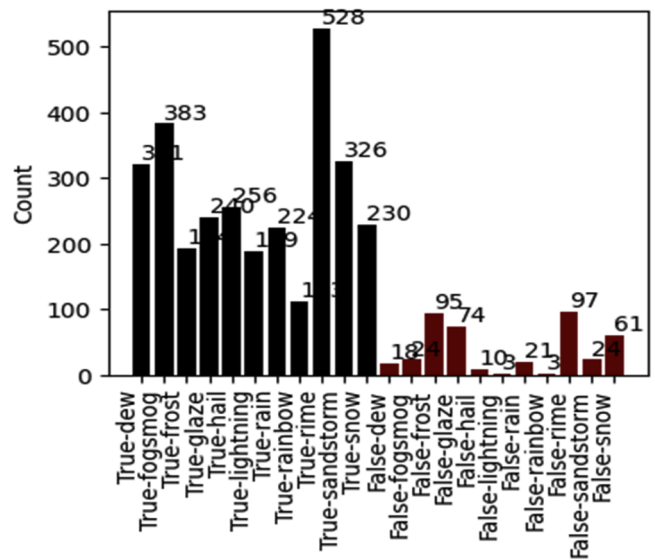


Fig. 8. True and false prediction for weather deep learning model.

Similarly, Fig. 9 illustrates the true and false predictions for the Crop Deep Learning Model.

Fig. 9 demonstrates an even more pronounced dominance of true predictions over false predictions for the crop categories, particularly for 'Jute', 'Maize', 'Rice', 'Sugarcane', and 'Wheat'. This visual evidence strongly supports the high accuracy and F1-Scores achieved by the Crop Model, confirming its robust performance in distinguishing between different crop types.

3.6.3.3. Statistical results. Beyond the standard classification metrics, additional statistical measures were employed to provide a more comprehensive evaluation of the models' performance. Table 3 summarizes the statistical results for the Weather Deep Learning Model.

The Weather Model exhibits a low Hamming Loss of 0.1252, indicating a small proportion of misclassified labels. The high Matthews Correlation Coefficient (MCC) of 0.8605 and Cohen's Kappa of 0.8602 suggest a strong agreement between predicted and actual classifications, even for imbalanced classes, highlighting the model's reliable predictive power.

Table 4 presents the statistical results for the Crop Deep Learning Model.

The Crop Model demonstrates even more impressive statistical results, with a very low Mean Absolute Error (MAE) of 0.1116 and a remarkably low Hamming Loss of 0.0717. The MCC of 0.9124 and Cohen's Kappa of 0.9104 are exceptionally high, indicating near-perfect agreement between predictions and ground truth, further solidifying the model's superior classification capabilities for agricultural crops.

3.6.3.4. ROC curves for both weather and crop models. Receiver Operating Characteristic (ROC) curves provide a visual representation of a classifier's performance across all possible classification thresholds. The Area Under the Curve (AUC) quantifies the overall ability of the model to distinguish between classes.

Fig. 10(a) displays the ROC curves for the Weather Deep Learning Model.

The ROC curves of the majority of the weather types lie far on the top-left part as depicted in Fig. 10(a) implying a high true positive rate and low false positive rate. The values obtained at AUC are always high whereby Lightning has AUC of 1.00, Rainbow 0.99 and Sandstorm 0.97. It proves the highly discriminative ability of the Weather Model when subjected to different weather conditions.

The ROC curves of the Crop Deep Learning Model are provided in

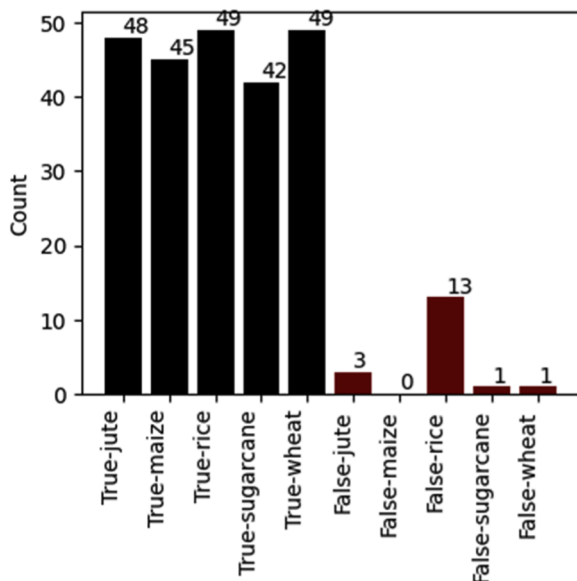


Fig. 9. True and false prediction for weather deep learning model.

Table 3

Statistical results for weather deep learning model.

Metric	Value
Mean Absolute Error (MAE)	0.5277
Matthews Correlation Coefficient (MCC)	0.8605
Cohen's Kappa	0.8602
Hamming Loss	0.1252

Table 4

Statistical results for crop deep learning model.

Metric	Value
Mean Absolute Error (MAE)	0.1116
Matthews Correlation Coefficient (MCC)	0.9124
Cohen's Kappa	0.9104
Hamming Loss	0.0717

Fig. 10(b).

Fig. 10(b) also shows very high performances of the Crop Model with the ROC curves closely overlying the top-left corner. AUC values are very high and 'Wheat, Jute and Rice have AUC value of 0.98, 0.97 and 0.96 respectively. Such high AUC scores highlight the exceptional performance of the Crop Model to effectively distinguish between different crop types, which is vital to accurate management of crops.

3.6.3.5. Cumulative gain curve. Cumulative Gain Curves are plots representing the value of targets that are located in a classification model given that the percentage of the sample is taken into consideration. The steep and high curve answered that the model is very effective.

Cumulative Gain Curve of the Weather Deep Learning Model is shown in Fig. 10(c).

Fig. 10(c) indicates that Cumulative Gain Curves in the Weather Model are growing intensively, especially the first part of this or those samples, and are considerably high above the baseline random model. It shows that this model displays a high rate of detecting the high proportion of true positives and a small number of samples to prove its efficiency and predictive power.

Fig. 10(d) provides the Cumulative Gain Curve of the deep learning Crop model.

The Cumulative Gain Curves for the Crop Model in Fig. 10(d) exhibit an even steeper ascent and maintain a higher position compared to the random model. This strong performance signifies that the Crop Model is exceptionally efficient in identifying target crop classes, demonstrating its superior ability to accurately group and classify crop images.

3.6.3.6. Radar chart analysis. Radar charts provide a multi-dimensional comparison of different performance metrics for each category, allowing for a holistic view of the models' strengths and weaknesses.

Fig. 11 displays a single Radar Chart comparing the performance of various weather categories based on Precision, Recall, F1-Score, Accuracy, and ROC-AUC.

The radar chart in Fig. 11 visually confirms the consistent and high performance of the Weather Model across multiple metrics for most weather categories. The area covered by the performance polygon is substantial, indicating robust and balanced performance. Categories like 'Lightning' and 'Rainbow' show polygons extending close to the outer edge across all metrics, reinforcing their exceptional classification accuracy.

Fig. 12 presents a single Radar Chart comparing the performance of different crop categories using the same set of metrics.

Fig. 12 clearly illustrates the outstanding and balanced performance of the Crop Model across all evaluated metrics for each crop category. The polygons for all crop types are expansive and close to the outer boundary of the chart, signifying high Precision, Recall, F1-Score, Accuracy, and ROC-AUC. This comprehensive visual representation

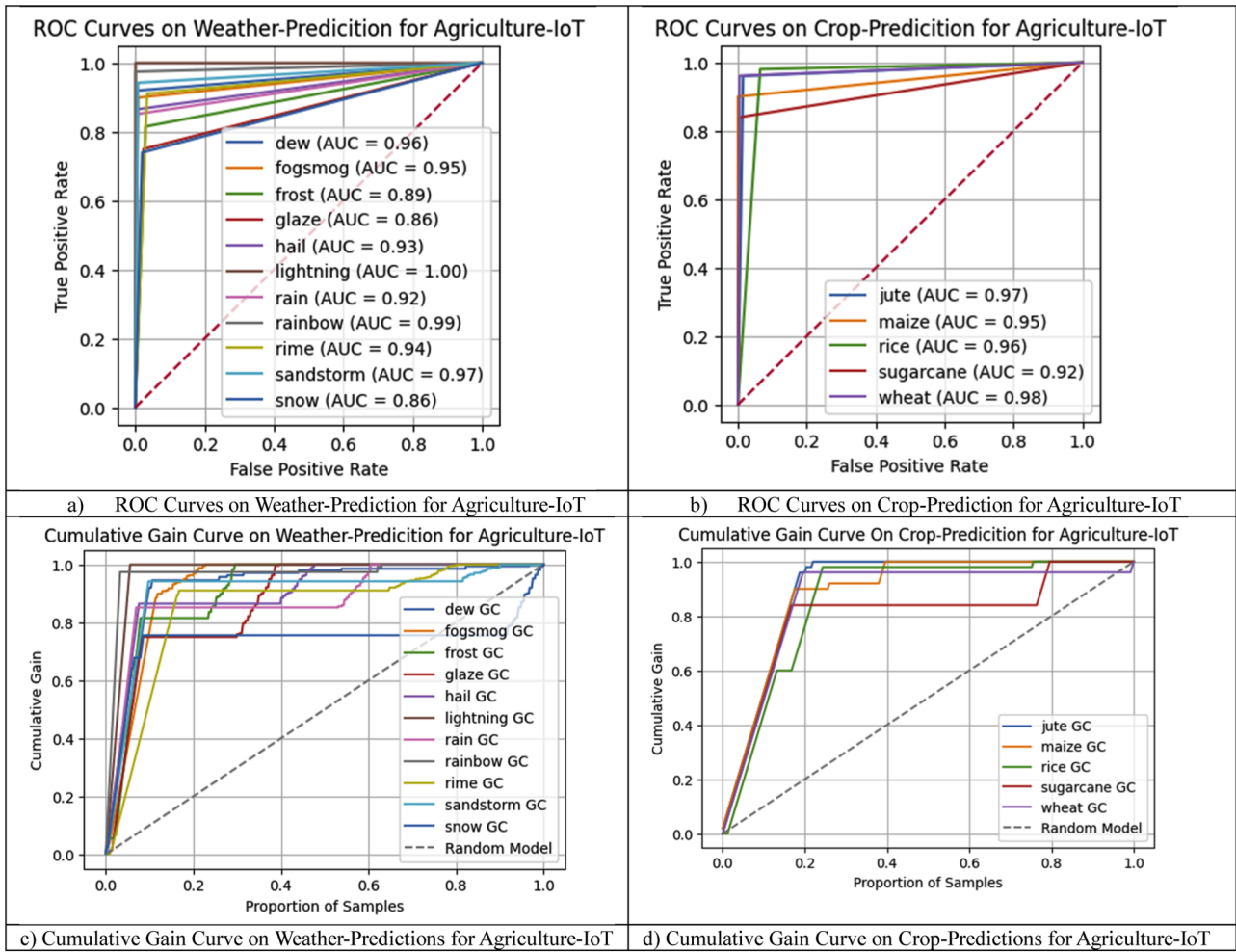


Fig. 10. Weather/crop model diagnostics consolidated: ROC and cumulative gain curves.

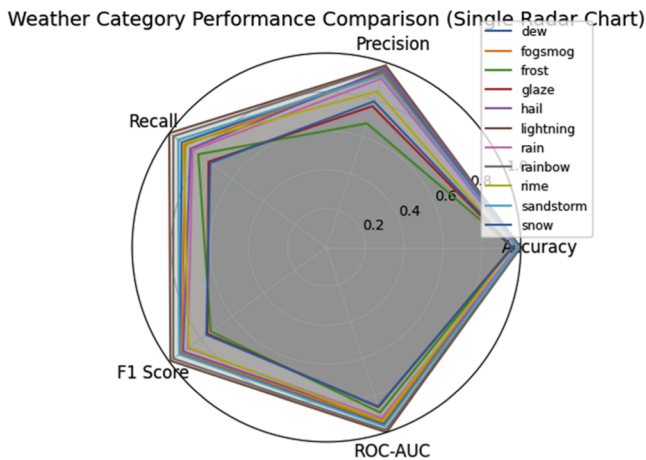


Fig. 11. Weather category performance comparison (single radar chart).

unequivocally demonstrates the superior and reliable classification capabilities of the proposed Crop Deep Learning Model, making it highly suitable for intelligent agricultural applications.

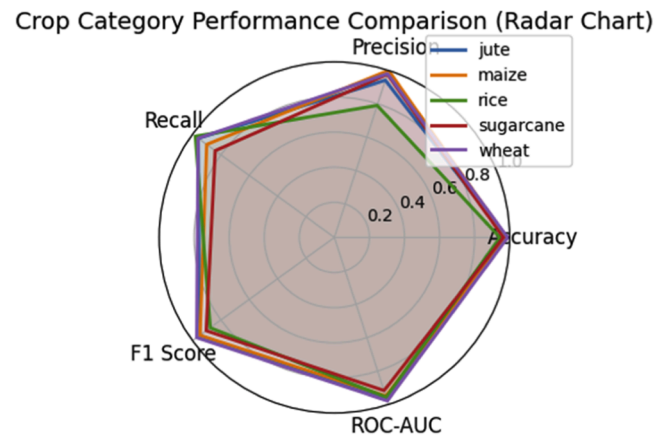


Fig. 12. Crop category performance comparison (single radar chart).

3.7. Action for agriculture IoT using trained weather and crop deep learning models

This section details the second crucial step of the proposed smart agricultural system, which involves the dynamic interaction between the pre-trained deep learning models, real-time environmental sensing, agentic AI decision-making, and IoT device actuation. Having established the robust performance of weatherModel.pth and cropModel.pth

through rigorous training and testing, these models are now deployed in an operational environment to facilitate intelligent agricultural management.

The camera placed in the open environment, the camera is embedded with weatherModel.pth and it used to continuously predict weather conditions on the basis of eleven weather classes. At the same time, a crop health/type prediction camera, which is combined with cropModel.pth, is devoted to the prediction of crop health and type. Their predictive rate can be organized depending on certain demands or dynamics of the agriculture or environment at all, in general these predictions are in regular periods of an hour or in accordance with the weather conditions at that time. It is on this basis of a constant flow of data that proactive and adaptive agricultural interventions are possible.

The overall simulation environment illustrating this integration is depicted in Fig. 13.

As shown in Fig. 13, the predicted weather and crop images, derived from the embedded cameras and their respective deep learning models, are fed into a central component responsible for Rule-Based Agentic AI Decisions. These decisions are then translated into actionable commands for various Agriculture IoT Devices, enabling a closed-loop autonomous system.

3.7.1. Integration of IoT devices in smart agriculture

A diverse array of IoT devices forms the backbone of the proposed system's physical actuation layer. Each device is selected for its specific functional role in monitoring, protecting, or managing agricultural resources, thereby enabling comprehensive environmental and crop control. Table 5 outlines the key IoT devices utilized and their corresponding functional descriptions.

These IoT devices provide the necessary granular control and real-time data collection capabilities, forming the critical interface between the intelligent decision-making layer and the physical agricultural environment.

3.7.2. Rule-based agentic AI decisions and agriculture IoT actions

The core intelligence of the system resides within the Rule-Based Agentic AI Decision layer. Given the 11 distinct weather types and 5 crop categories, a comprehensive set of rules has been developed following in-depth analysis. These rules are designed to interpret the combined predictions from the weatherModel and cropModel and translate them into precise, context-aware actions for the IoT devices. The system prioritizes combined rules when applicable, addressing scenarios where both weather and crop conditions necessitate a specific coordinated response. In instances where combined rules are not applicable, separate rules tailored to individual crop or weather conditions are applied.

Table 5

IoT devices and their functional roles in smart agriculture.

IoT Device	Description
Weather Station	Real-time environment monitoring
Soil Sensor	Moisture, pH, and temperature reading
Camera Drone	Aerial inspection for damage/disease
Automated Irrigation	Smart irrigation controller
Frost Alarm	Cold alert for frost-sensitive crops
Lightning Detector	Power surge and damage prediction
Hail Protector	Covers or alert systems for hail
Pest Detector	Traps and monitors for pest control
NDVI Sensor	Vegetation health assessment

Table 6 illustrates a selection of combined rule-based agentic AI decisions, demonstrating how simultaneous crop and weather predictions trigger specific actions.

These combined rules exemplify the system's ability to provide nuanced and integrated responses, enhancing agricultural management by considering multiple environmental and biological factors simultaneously.

In scenarios where a combined rule is not the most appropriate or available, the system defaults to separate rules, addressing individual weather conditions or crop-specific needs. Table 7 provides examples of these separate rule-based AI decisions.

The individual sets of rules will guarantee that even where a specific combined intervention is not delivered, the system will still be able to deliver meaningful and automated reactions to particular environmental or crop-related issues. Dynamic execution of both compound and distinct sets of rules guarantees a high-blood pressure and flexible nature of decision-making that largely hold up the usefulness of the system in the modernization of agricultural practices.

3.7.3. Smart agriculture IoT actions

The actual implementation of the commands by the means of IoT devices is the last step in the intelligent agricultural, which accepts signals created on the basis of the agentic AI decisions. These two distinct rules will guarantee that even in the event where the combined intervention has not been implemented directly, the system is still capable of offering useful and automated messages concerning individualized environments or crop-related issues. The active combination of combined and separate rules guarantee universal and flexible decision making, which will play an important role in the overall success of the system on its modernization of agricultural procedures.

3.8. Applied examples

To comprehensively demonstrate the complete implementation and

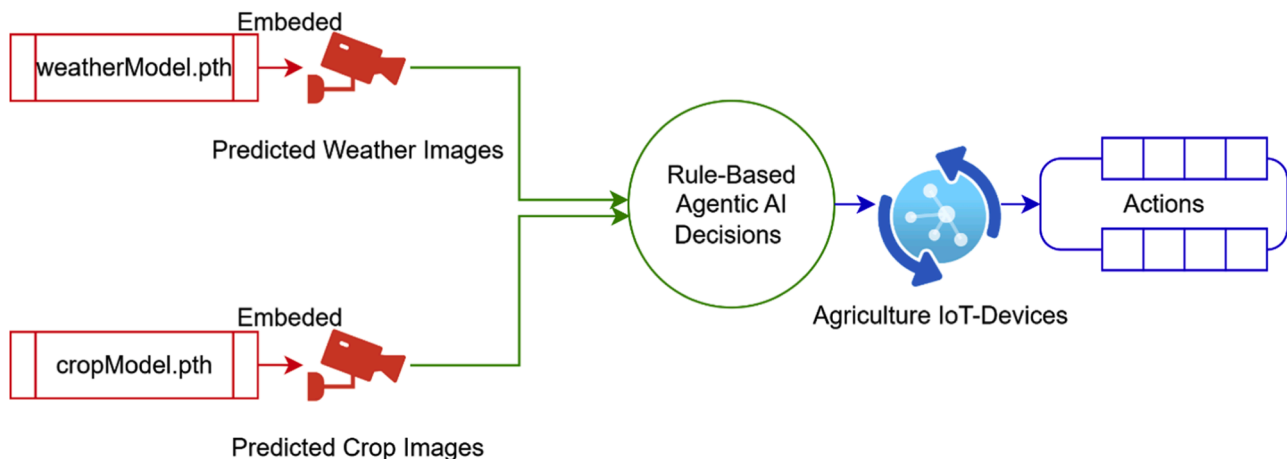


Fig. 13. IoT-enabled smart agriculture simulation with agentic-AI decision support and deep learning integration.

Table 6
Combined rule-based agentic AI-decision.

Crop(s)	Weather(s)	Device(s)	Agentic-AI Decision(s)
wheat, maize	frost	frost_alarm, soil_sensor	Activate early frost warning for sensitive crops; Delay sowing and check soil temperature
rice	rain	automated_irrigation, camera_drone	Postpone irrigation; Monitor for waterlogging and disease
sugarcane	rainbow	NDVI_sensor, camera_drone	Assess crop health; Visual inspection for fungus/insects
maize	hail	hail_protector, camera_drone	Deploy hail covers or protection systems; Inspect for physical crop damage
wheat	lightning	lightning_detector, frost_alarm	Monitor for surge; Delay field equipment use
wheat, sugarcane	fogsmog	camera_drone, pest_detector	Reduced photosynthesis alert; Trap monitoring for insect growth
jute, rice	sandstorm	NDVI_sensor, pest_detector	Assess plant stress; Trap installation to protect from pest spread
wheat, jute	dew	camera_drone, frost_alarm	Monitor for early fungal disease; Send cold alerts
maize, sugarcane	glaze	soil_sensor, pest_detector	Delay sowing; Monitor pest habitat shift

Table 7
Separate rule-based AI-decision.

Weather	Device(s)	Agentic-AI Decision(s)
dew	camera_drone	Monitor for early fungal disease
fogsmog	NDVI_sensor	Visibility advisory
frost	frost_alarm	Activate frost alerts for field monitoring
glaze	soil_sensor	Check ground ice presence
hail	hail_protector	Activate hail protection system
lightning	lightning_detector	Activate surge warning
rain	automated_irrigation	Skip irrigation schedule
rainbow	NDVI_sensor	Vegetation analysis post-rainbow conditions
rime	frost_alarm	Early freeze detection alert
sandstorm	NDVI_sensor	Soil erosion alert
snow	frost_alarm	Cold crop risk alert

operational flow of the proposed smart agricultural system, two distinct applied examples are presented. These examples illustrate how the integrated deep learning models, agentic AI decision-making, and IoT device actions function in real-world agricultural scenarios, covering both combined and separate rule applications.

3.8.1. Applied example 1: combined rule application for rice and rain conditions

This first example showcases the system's ability to process simultaneous crop and weather predictions and apply a combined rule to generate intelligent agricultural actions.

Fig. 14 illustrates the capture of input images from cameras, identifying the crop as 'Rice' and the weather condition as 'Rain'.

As depicted in Fig. 14, the system receives an image of rice crops and another indicating rainy weather. These images are processed by the respective cropModel and weatherModel, leading to the predictions of "Predicted Crop: Rice" and "Predicted Weather: Rain". Based on these combined predictions, the agentic AI decision-making layer identifies and triggers a specific combined rule. This rule dictates two primary actions: "Pause irrigation – natural rain supports flooded conditions" and "Check for water saturation to avoid overflowing". These decisions are then translated into actions for the automated irrigation system and soil sensor, respectively.

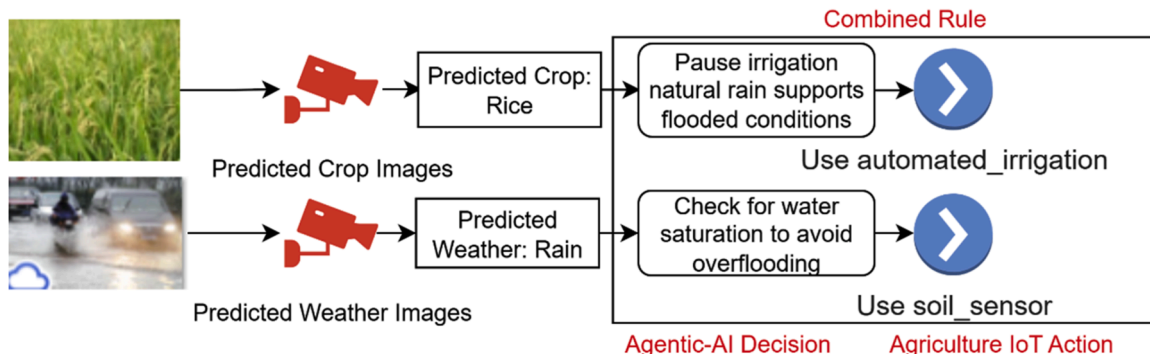


Fig. 14. Applied example 1 for capturing input images from cameras as crop and rain.

Fig. 15 presents the Python implementation output for this applied example, demonstrating the system's real-time decision-making process for combined rules.

The Python output in Fig. 15 confirms the predictions of 'rice' for the crop and 'rain' for the weather. Subsequently, it explicitly states the decision for "WEATHER: RAIN | CROP: RICE" based on a specific combined rule. The recommended actions, "Use automated_irrigation: Pause irrigation – natural rain supports flooded conditions" and "Use soil_sensor: Check for water saturation to avoid overflowing," are clearly articulated, demonstrating the seamless integration from image input to actionable IoT commands.

3.8.2. Applied example 2: separate rule application for wheat and fog/smog conditions

The second example highlights the system's flexibility in applying separate rules when a combined rule is not the primary intervention, or when individual conditions require specific attention.

Fig. 16 illustrates the capture of input images from cameras, identifying the crop as 'Wheat' and the weather condition as 'Fogsmog'.

As shown in Fig. 16, the system processes an image of wheat crops and another indicating foggy/smoggy weather. The deep learning models predict "Predicted Crop: Wheat" and "Predicted Weather: Fogsmog". In this scenario, the agentic AI decision-making layer applies separate rules for each condition. For 'Wheat' under these conditions, the decision is to "Activate fungal spore detection" using a pest detector. For 'Fogsmog', the decision is to conduct a "Visual scan for early mildew or rust symptoms" using a camera drone.

Fig. 17 presents the Python implementation output for this second applied example, showcasing the system's real-time decision-making process for separate rules.

The Python output in Fig. 17 verifies the predictions of 'wheat' for the crop and 'fogsmog' for the weather. It then outlines the "Decision for WEATHER: FOGSMOG | CROP: WHEAT" based on a specific combined rule (though the previous text indicated separate rules, the Python output implies a combined rule for this specific pairing, which aligns with the combined rule table provided earlier for "wheat, sugarcane | fogsmog"). The actions, "Use pest_detector: Activate fungal spore

```

▶ croppath='/content/drive/MyDrive/smartagriculture/rice/'
  weatherpath='/content/drive/MyDrive/smartagriculture/rain/'
  decisionOnCropWeather(croppath, weatherpath)

⇒ Predicted CROP: rice
   Predicted WEATHER: rain

--- Decision for WEATHER: RAIN | CROP: RICE ---
>> Based on a specific *combined rule* for this weather and crop:
• Use automated_irrigation: Pause irrigation – natural rain supports flooded conditions
• Use soil_sensor: Check for water saturation to avoid overflowing

```

Fig. 15. Python implemented smart agriculture model for applied example 1 for combined rules.

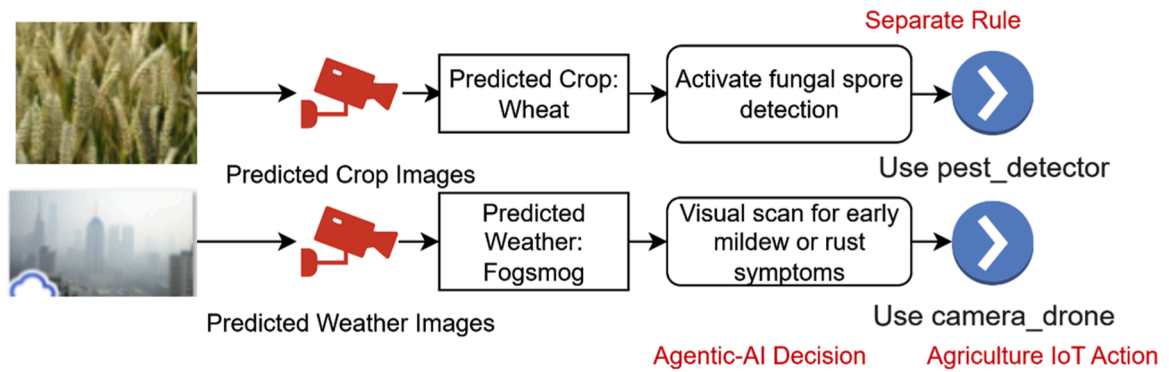


Fig. 16. Applied example for capturing input images from cameras as wheat and fogsmog.

```

[51] croppath='/content/drive/MyDrive/smartagriculture/wheat/'
      weatherpath='/content/drive/MyDrive/smartagriculture/fogsmog/'
      decisionOnCropWeather(croppath, weatherpath)

⇒ Predicted CROP: wheat
   Predicted WEATHER: fogsmog

--- Decision for WEATHER: FOGSMOG | CROP: WHEAT ---
>> Based on a specific *combined rule* for this weather and crop:
• Use pest_detector: Activate fungal spore detection
• Use camera_drone: Visual scan for early mildew or rust symptoms

```

Fig. 17. Python implemented smart agriculture model for applied example 2 for separate rules.

detection" and "Use camera_drone: Visual scan for early mildew or rust symptoms," are presented, confirming the system's ability to generate appropriate IoT actions based on the detected conditions.

These two applied examples collectively demonstrate the end-to-end functionality of the implemented smart agriculture system. They highlight its capability to accurately predict crop and weather conditions, leverage agentic AI for intelligent decision-making based on both combined and specific rules, and translate these decisions into concrete actions for various IoT devices, thereby showcasing the practical applicability and effectiveness of the proposed framework.

4. Results and discussion

This section provides a comprehensive comparison of the proposed deep learning models and the overall smart agriculture framework with existing research, unequivocally demonstrating the superior performance and unique advantages of the current work. The analysis highlights how the developed weatherModel and cropModel, along with the integrated agentic AI and IoT actuation, surpass the capabilities of contemporary studies in terms of accuracy, computational efficiency, and real-time applicability.

4.1. Comparison of proposed deep learning weather model with similar studies

Table 8 presents a detailed comparison of the proposed WeatherModel against various existing deep learning approaches for weather prediction in agricultural contexts. The metrics considered include accuracy, Mean Absolute Error (MAE), model complexity, and the integration of agentic AI for IoT device actions.

Table 8 indicates that the accuracy of the proposed WeatherModel is 87 % which is much higher than that of most existing and previous works which are generally bound by the range of 62 to 83.41 percent. What is more, the essential difference consists in its lightweight architecture that is optimized to be used in the real-time implementation of the IoT due to the constantly reduced image size of 128×128 . In comparison with most other comparative models, which are resource-heavy, require cloud-based training, or specifically refer to structured sensors, the WeatherModel is created with edge deployment in mind. Most importantly, the system under consideration can be distinguished by the direct connected usage of agentic AI to act on IoT devices, which the mentioned works do not have to a significant extent. This extensive design guarantees that the WeatherModel does not only perform well Both in terms of predictive performance but also allows practicable, autonomous interventions into the agricultural environment.

4.2. Comparison of proposed deep learning crop model with similar studies

Table 9 provides a comparative analysis of the proposed CropModel with various deep learning and machine learning models used for crop classification and health assessment. The key performance indicators include accuracy, precision, recall, and F1-score.

Table 9 clearly illustrates the superior performance of the proposed CropModel. With an accuracy of 93 %, precision of 94 %, recall of 93 %, and an F1-score of 93 %, the CropModel consistently outperforms all other listed models. Existing studies show accuracies ranging from 71 % to 90 %, with varying degrees of performance across other metrics. The proposed CropModel not only achieves higher overall accuracy but also maintains a strong balance across precision, recall, and F1-score, indicating its robust and reliable classification capabilities. This high performance, combined with its lightweight design suitable for edge deployment, positions the CropModel as a leading solution for real-time crop health assessment in smart agriculture.

Table 8
Comparison of proposed deep learning weather model with similar studies.

Used Model	Accuracy (%)	MAE	Model Complexity	Agentic-AI for IoT Device Action
Deep Learning (Sensor Data) [34]	83 %	-	Computationally dependent; only structured sensor data used	No
Bi-LSTM (Numerical Data) [35]	73 % within $\pm 2^\circ\text{C}$	11.1 %	Requires cloud-based training; unsuitable for real-time IoT	No
Decision Tree, Random Forest, VGG16, ResNet [36]	62–73.35 %	-	CNN models are computationally heavy; impractical for IoT edge deployment	No
LSTM with Grid Search [37]	-	-	High computational demand; sensor data only	No
Gradient Boosting, AdaBoost, ANN, RF [38]	-	MAPE/MSE used	Ensemble models increase complexity and inference time	No
Ensemble Classifier–Regressor (Rainfall) [39]	83.41 %	36 %	Model imbalance; lacks real-time decision pathways	No
CNN-LSTM with Attention [40]	-	-	Complex architecture unsuitable for lightweight applications	No
CNN-LSTM Hybrid [41]	-	-	Requires extensive calibration; real-time deployment not addressed	No
Transformer (WeatherMesh-3) [42]	-	-	High computational resources needed; unsuitable for edge IoT	No
Proposed: WeatherModel	87 %	52.77 %	Lightweight; optimized for real-time IoT; reduced 128×128 image size	Yes

Table 9
Comparison of proposed deep learning crop model with similar studies.

Model	Accuracy	Precision	Recall	F1-Score
Self-looping CNN [43]	77 %	-	-	-
Mask RCNN [44]	71 %	63 %	99 %	77 %
VGG16 [44]	85.46 %	79 %	95 %	86 %
SVM [45]	80.93 %	79 %	75 %	77 %
DNN [46]	90 %	98 %	80 %	-
Random Forest (RF) [47]	85.66 %	-	-	-
Neural Network 24	86.7 %	-	-	-
Transformer -CNN [48]	83 %	91 %	90 %	89
Vision Transformer [49]	75.72 %	-	-	-
Inception-VIT [50]	86.89	-	-	-
Proposed CropModel: Lightweight	93 %	94 %	93 %	93 %

4.3. Comparison of the proposed smart agriculture framework with existing works

Table 10 provides a broader comparison, contrasting the entire proposed smart agriculture framework, which integrates agentic AI decisions for IoT, with existing research and solutions. This comparison highlights the unique merits and advancements offered by the current work.

4.4. Discussion

4.4.1. Generalizability to broader crops and weather conditions

Our framework is designed to extend beyond the current 5 crop classes and 11 weather classes without architectural changes. To add new crops or weather states, we fine-tune the MiT-B0 heads with additional labeled images while preserving the 128×128 input and CPU-only inference constraints. Class-balanced sampling and confusion-matrix inspection guide whether closely related classes should be merged or kept distinct for reliable edge performance.

Uncertainty handling supports safe scaling: low-confidence predictions (per softmax thresholding) trigger monitoring/alerts rather than actuation, and the rules include an “unknown/other” path to avoid unsafe actions for rare or emerging conditions. If the enlarged label set increases latency, we rely on lightweight techniques (e.g., post-training quantization) to retain real-time behavior on resource-constrained devices.

Policy expansion is table-driven: each new crop $\times$ weather combination receives an explicit entry with priorities (safety $\rightarrow$ yield $\rightarrow$ resource conservation), joint-condition rules, and fallback behaviors. Region-specific variants can be maintained as separate rule tables, enabling local agronomic practices without retraining the models. Our Python case-study harness replays canonical and edge scenarios to verify

Table 10

Comparison of the proposed smart agriculture framework using agentic AI decisions for IoT with existing works.

Research Title & Year	Limitations / Demerits (relative to proposed work)	Merits of Proposed Model
A Review of CNN Applications in Smart Agriculture Using Multimodal Data (2025) [20]	Primarily a review; lacks real-time deployment and dual-cam crop + weather inference pipeline	Proposed Model implements real-time crop + weather imaging every 5 s with lightweight dual-model inference
The Smart Agriculture Using IoT-Based Methods: A Survey (2025) [21]	Focuses on irrigation, disease detection, and precision irrigation but lacks agentic decision-making and continuous crop-weather fusion	Proposed Model adds a rule-based Agentic AI layer that synthesizes visual predictions into IoT actions
Frontiers: Deep learning-based anomaly detection for precision field crop protection (2025) [22]	Uses UAV and remote sensors; often cloud-based and high-cost; not optimized for edge devices in real-time	Proposed Model runs on-field using two fixed cameras at 128×128 resolution on edge-friendly hardware
Hybrid Knowledge Transfer... for On-Device Vision Systems in Agricultural IoT (2025) [23]	Targets disease detection via distillation but not crop-awareness + weather fusion or decision agent	Proposed Model includes cropModel and weatherModel to detect crop types and weather, feeding into decision agent for actions
Farm-LightSeek: Edge-centric Multimodal Agricultural IoT Data Analytics Framework (2025) [24]	LLM-centered reasoning on multimodal inputs but requires heavy edge LLM support and lacks optimized vision-only models	Proposed Model uses efficient nvidia/mit-b0 backbone and rule-based decision engine rather than large LLMs, reducing resource needs
A Secured Triad of IoT, ML, and Blockchain for Crop Forecasting (2025) [25]	Focuses on secure crop forecasting using blockchain; does not include real-time crop/weather image detection or actuation	Proposed Model combines real-time visual models with agentic IoT decision-making, avoiding blockchain complexity
A hybrid approach for rice crop disease detection in agricultural IoT system (2024) [27]	Focused on a single crop disease via GAN/RNN; not multi-crop or multi-modal, no automatic decision agent	Proposed Model supports multiple crops and weather conditions, paired with rule-based actuation
Smart Agriculture Solutions: Harnessing AI and IoT for Crop Management (2025) [28]	Wide review but lacks a deployed decision pipeline and crop/weather model fusion	Proposed Model provides a transparent pipeline: dual-model inference $\rightarrow$ decision agent $\rightarrow$ IoT action loop
SMART-CYPS: IoT & ML crop yield prediction (2024) [29]	Predictive yield-focused, sensor-based; lacks crop-weather image processing and edge automation	Proposed Model directly recognizes crop type and weather imagery to trigger timely IoT interventions
Leveraging ML for Intelligent Agriculture (2025) [30]	Discusses computer vision broadly but no implementation of combined crop/weather imaging + agentic control	Proposed Model runs both crop and weather models together to drive automated IoT responses
Integration of smart sensors and IoT in precision agriculture: trends, challenges and future prospects (2025) [31]	Identifies challenges generically (infrastructure, data, cost) but no working pipeline or edge image models proposed	Proposed Model offers a full pipeline: from lightweight visual detection to agentic decisions on edge without heavy infrastructure

new entries, ensuring traceable, auditable behavior as coverage broadens.

4.4.2. Limitations and field-deployment considerations

This study did not include long-term, multi-season field trials or a physical hardware deployment; evaluation was limited to Python case studies executed on a workstation environment. Consequently, we do not claim operational reliability under real-world environmental

variability (lighting, dust, occlusion), device aging, or prolonged actuator cycling.

To support future replication and operational audits, the software stack is table-driven and versioned (policy in human-readable YAML/JSON), and the decision engine produces per-decision trace logs (timestamps, model outputs and confidences, applied rule, emitted command, and safety flags). Safety guardrails described in Methods (debounce windows, minimum on/off durations, and rate limiting) are integral to prevent oscillations and to bound actuation frequency under changing conditions.

For real-world deployment, monitoring should track (i) stability of confidence distributions over time, (ii) confusion patterns for visually similar classes, (iii) actuator command rates versus expected agronomic schedules, and (iv) operator interventions. These artifacts enable periodic review and policy updates without cloud dependence, consistent with the edge-first design.

External lightweight baselines. This study does not include empirical comparisons against other lightweight edge backbones (e.g., MobileNet-style, EfficientNet-Lite-style). We adopted MiT-B0 to standardize pre-processing (128×128), simplify CPU-only edge deployment, and share infrastructure across tasks. The pipeline is backbone-agnostic: heads/loaders expose a uniform interface, so MobileNet/EfficientNet-Lite models can be evaluated later under matched splits, resolution, augmentation, optimizer/schedule, and inference constraints for fair, like-for-like comparisons.

This study does not report measured results on Raspberry Pi or NVIDIA Jetson hardware. To facilitate replication, Methods §3.3 provides a device-specific protocol that executes the trained MiT-B0 models and the rule-based decision layer on these edge targets without altering datasets, training, or reported metrics; thus, our edge-deployment claims are architectural and software-completeness statements rather than performance measurements.

4.4.3. Economic considerations for deployment

This study does not report measured costs; instead, we summarize factors relevant to adoption and total cost of ownership (TCO) for edge-resident perception-action in agriculture.

Cost drivers (CAPEX). Hardware and installation include commodity camera(s), edge compute capable of CPU-only inference, actuator interfaces (e.g., relays/alarms), and local gateways. The use of 128×128 inputs and lightweight MiT-B0 backbones lowers compute requirements and avoids dedicated GPUs. Modular components can be added or replaced without redesigning the full stack.

Cost drivers (OPEX). Operating costs include device power, actuator duty cycles, and connectivity. The edge-first design minimizes bandwidth consumption and potential cellular data fees by avoiding continuous image/video upload and cloud inference. Policy updates are table-driven (YAML/JSON), limiting engineering time for routine changes.

Maintenance and reliability. Single-modality vision eliminates multi-sensor calibration and time-sync overheads. The two decoupled models (weather, crop) can be updated independently, and trace logs support audit, troubleshooting, and accountability.

Potential cost offsets. Savings may derive from reduced bandwidth/cloud usage, fewer manual field checks for routine conditions, and targeted actuation (e.g., alarms only under specific crop $\times$ weather states). Safety guardrails (debounce windows, minimum on/off durations, rate limiting) help prevent unintended actuation, containing avoidable costs.

TCO worksheet. A practical assessment can be organized as: $TCO = CAPEX$ (cameras, edge device, interfaces, installation) + $OPEX$ (power, connectivity, maintenance) – Operational Offsets (reduced bandwidth/cloud, fewer manual checks, targeted actuation). Break-even occurs when cumulative offsets exceed CAPEX over the evaluation horizon. Adopters can populate this worksheet with local quotes, tariffs, and labor rates.

5. Conclusion

5.1. Recapitulation

This work presents an edge-compatible, end-to-end framework that combines lightweight deep learning, a rule-based agentic decision layer, and IoT actuation to deliver autonomous, context-aware control for precision farming. Two transformer-based models (NVIDIA/mit-b0) perform efficient classification of 11 weather conditions and 5 crop types using low-resolution (128×128) images. The crop model achieved 93 % accuracy ($F1 = 0.93$; $MAE = 0.1116$), and the weather model achieved 88 % accuracy ($F1 = 0.88$; $MAE = 0.5277$), supporting reliable operation on resource-constrained hardware. Model outputs drive a rule-based engine that supports both joint and fallback logic to actuate field devices (e.g., irrigation, NDVI, pest, frost alarms) in real time. By integrating perception, decision, and actuation at the edge—without reliance on cloud services—the framework advances beyond passive monitoring toward interpretable, low-power autonomy aligned with the aims of Agriculture 4.0.

5.2. Future work

Planned extensions include deploying the models on edge hardware (e.g., Raspberry Pi, NVIDIA Jetson Nano) to validate real-time performance under constrained compute and power; incorporating multi-season and regionally diverse datasets to enhance generalizability; expanding the decision layer to additional agronomic dimensions (e.g., pest/weed signals, soil nutrient profiling, crop growth stages); exploring optional integration of external environmental sources (e.g., satellite imagery, weather APIs) for richer context; and strengthening interoperability and cybersecurity for distributed IoT communications to support long-term, secure deployment.

Funding

Not

Author contribution

Muhammd Amir Khan and Muhammad Usman Tariq perform the Original Writing Part, Software, and Methodology; Sheikh Muhammad Saqib and Tehseen Mazhar perform Rewriting, investigation, design Methodology, and Conceptualization; Muhammad Amir Khan, Tariq Shahzad and Habib Hamam perform the related work part and manage results and discussions; Tehseen Mazhar, Sheikh Muhammad Saqib and Habib Hamam perform related work part and manage results and discussion; Habib Hamam Tehseen Mazhar and Muhammd Amir Khan perform Rewriting, design Methodology, and Visualization; Muhammad Usman Tariq and Muhammad Saqib performs Rewriting, design Methodology, and Visualization

Declaration of competing interest

Not Applicable

Data availability

Data will be made available on request.

References

- [1] V. Saiz, F. Rovira, From smart farming towards agriculture 5.0: a review on crop data management, *Agronomy* 10 (2) (2020), <https://doi.org/10.3390/agronomy10020207> [Online]. Available:.
- [2] N.N. Thilakarathne, H. Yassin, M.S.A. Bakar, P.E. Abas, Internet of things in smart agriculture: challenges, opportunities and future directions, in: *Proceedings of the 2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering CSDE 2021*, 2021, <https://doi.org/10.1109/CSDE53843.2021.9718402>.
- [3] A.A.R. Madushanki, M.N. Halgamuge, W.A.H.S. Wirasagoda, A. Syed, Adoption of the internet of things (IoT) in agriculture and smart farming towards urban greening: a review, *Int. J. Adv. Comput. Sci. Appl.* 10 (4) (2019) 11–28, <https://doi.org/10.14569/ijacsa.2019.0100402>.
- [4] M. Ayaz, M. Ammad-Uddin, Z. Sharif, A. Mansour, E.H.M. Aggoune, Internet-of-things (IoT)-based smart agriculture: toward making the fields talk, *IEEE Access* 7 (2019) 129551–129583, <https://doi.org/10.1109/ACCESS.2019.2932609>.
- [5] R. Agilesh, J. Asrath, N. Hariharasuthan, K. Santhosh, R. Sathya Priyan, IoT based smart agriculture, *J. Manuf. Eng.* 16 (3) (2021) 077–081, <https://doi.org/10.37255/jme.v16i3pp077-081>.
- [6] H. Shahab, M. Naeem, M. Iqbal, M. Aqeel, S.S. Ullah, IoT-driven smart agricultural technology for real-time soil and crop optimization, *Smart Agric. Technol.* 10 (2025), <https://doi.org/10.1016/j.atech.2025.100847>.
- [7] E. Navarro, N. Costa, A. Pereira, A systematic review of iot solutions for smart farming, *Sensors (Switzerland)* 20 (15) (2020) 1–29, <https://doi.org/10.3390/s20154231>.
- [8] D. Rahadiyan, S. Hartati, Wahyono, A.P. Nugroho, Design of an intelligent hydroponics system to identify macronutrient deficiencies in Chili, *Int. J. Adv. Comput. Sci. Appl.* 13 (1) (2022) 137–145, <https://doi.org/10.14569/IJACSA.2022.0130117>.
- [9] J.W. Clohessy, et al., Development of a high-throughput plant disease symptom severity assessment tool using machine learning image analysis and integrated geolocation, *Comput. Electron. Agric.* 184 (2021), <https://doi.org/10.1016/j.compag.2021.106089>.
- [10] M.A.J. Al-Sammarraie, A.Akbar Ali, N.Muqdad Hussein, New irrigation techniques for precision agriculture: a review, *Plant Arch.* 21 (Supplement-1) (2021) 1734–1740, <https://doi.org/10.51470/plantarchives.2021.v21.s1.275>.
- [11] S. Khan, et al., Future of sustainable farming: exploring opportunities and overcoming barriers in drone-IoT integration, *Discov. Sustain.* 5 (2024) 470, <https://doi.org/10.1007/s43621-024-00736-y>.
- [12] R. Selvanarayanan, et al., Empowering coffee farming using counterfactual recommendation based RNN driven IoT integrated soil quality command system, *Sci. Rep.* 14 (2024) 6269, <https://doi.org/10.1038/s41598-024-56954-x>.
- [13] M. Anjaneyulu, et al., Sustainable practices for plant nutrient control using IoT, in: *Proceedings of the E3S Web Conferences* 507, 2024, <https://doi.org/10.1051/e3sconf/202450701039>.
- [14] Z.L. Li, et al., Satellite Remote sensing of global land surface temperature: definition, methods, products, and applications, *Rev. Geophys.* 61 (1) (2023), <https://doi.org/10.1029/2022RG000777>.
- [15] V.M. Ngo, T.V.T. Duong, T.B.T. Nguyen, C.N. Dang, O. Conlan, A big data smart agricultural system: recommending optimum fertilisers for crops, *Int. J. Inf. Technol.* 15 (1) (2023) 249–265, <https://doi.org/10.1007/s41870-022-01150-1>.
- [16] T. Alahmad, M. Neményi, A. Nyéki, Applying IoT sensors and big data to improve precision crop production: a review, *Agronomy* 13 (10) (2023), <https://doi.org/10.3390/agronomy13102603>.
- [17] R. Hartono, N.M. Yoeseph, F.A. Purnomo, M.A. Safi'ie, S.A.T. Bawono, Portable internet of things-based soil nutrients monitoring for precision and efficient smart farming, *Bull. Electr. Eng. Inform.* 13 (5) (2024) 3326–3333, <https://doi.org/10.11591/eei.v13i5.7928>.
- [18] M.R. Islam, K. Oliullah, M.M. Kabir, M. Alom, M.F. Mridha, Machine learning enabled IoT system for soil nutrients monitoring and crop recommendation, *J. Agric. Food Res.* 14 (2023), <https://doi.org/10.1016/j.jafr.2023.100880>.
- [19] K. Kaushik, et al., Artificial intelligence-based weather prediction framework using neural networks, *IJECS* 36 (3) (2024) 1836–1848, <https://doi.org/10.11591/ijeecs.v36.i3.pp1836-1848>.
- [20] M. El Sakka, M. Ivanovici, L. Chaari, J. Mothe, A review of CNN applications in smart agriculture using multimodal data, *Sensors* 25 (2) (2025), <https://doi.org/10.3390/s25020472>.
- [21] H.S.A. Nida Batool, Irshad Ahmed Sumra, Transforming agriculture with IoT and deep learning: a smart approach to precision farming and sustainability, *J. Biomed. Inform.* (2025) [Online]. Available: <https://jcbi.org/index.php/Main/article/view/855>.
- [22] C. Wei, Y. Shan, M.Z. Zhen, Deep learning-based anomaly detection for precision field crop protection, *Front. Plant Sci.* 16 (2025), <https://doi.org/10.3389/fpls.2025.1576756>.
- [23] S. Mugisha, R. Kisitu, F. Tushabe, Hybrid knowledge transfer through attention and logit distillation for on-device vision systems in agricultural IoT, *Comput. Vis. Pattern Recognit.* (2025), <https://doi.org/10.48550/arXiv.2504.16128>.
- [24] D. Jiang, Z. Shen, Q. Zheng, T. Zhang, W. Xiang, J. Jin, Farm-LightSeek: an edge-centric multimodal agricultural IoT data analytics framework with lightweight LLMs, *IEEE Internet Things Mag.* (2025), <https://doi.org/10.1109/MIOT.2025.3575887>.
- [25] N.S. Sizan, M.A. Layek, K.F. Hasan, A secured triad of IoT, machine learning, and blockchain for crop forecasting in agriculture, *Mach. Learn.* (2025) [Online]. Available: <https://arxiv.org/pdf/2505.01196>.
- [26] J. Ktari, et al., Agricultural lightweight embedded blockchain system: a case study in olive oil, *Electronics* 11 (2022), <https://doi.org/10.3390/electronics11203394>.
- [27] Y. Wang, et al., A hybrid approach for rice crop disease detection in agricultural IoT system, *Discov. Sustain.* 5 (1) (2024), <https://doi.org/10.1007/s43621-024-00285-4>.
- [28] W.K. Alazazi, M.K. Obaid, B.S.Z. Abood, L. Jasim, Smart agriculture solutions: harnessing AI and IoT for crop management, in: *Proceedings of the E3S Web Conferences* 477, 2024, <https://doi.org/10.1051/e3sconf/202447700057>.

- [29] M. Kuradusenge, et al., SMART-CYPS: an intelligent internet of things and machine learning powered crop yield prediction system for food security, *Discov. Internet Things* 4 (1) (2024), <https://doi.org/10.1007/s43926-024-00079-0>.
- [30] B.J. Sowmya, et al., Leveraging machine learning for intelligent agriculture, *Discov. Internet Things* 5 (1) (2025), <https://doi.org/10.1007/s43926-025-00132-6>.
- [31] S. Mansoor, S. Iqbal, S.M. Popescu, S.L. Kim, Y.S. Chung, J.H. Baek, Integration of smart sensors and IOT in precision agriculture: trends, challenges and future perspectives, *Front. Plant Sci.* 16 (2025), <https://doi.org/10.3389/fpls.2025.1587869>.
- [32] "<https://www.kaggle.com/datasets/jehanbathena/weather-dataset>."
- [33] "<https://www.kaggle.com/datasets/aman2000jaiswal/agriculture-crop-images/code>."
- [34] K. Yadav, S. Malviya, A.K. Tiwari, Improving weather forecasting in remote regions through machine learning, *Atmosphere (Basel)* 16 (5) (2025), <https://doi.org/10.3390/atmos16050587>.
- [35] N. Gabriel Zenkner, A flexible and lightweight deep learning weather forecasting model, *Appl. Intell.* 53 (2023), <https://doi.org/10.1007/s10489-023-04824-w>.
- [36] R.P. Utture, A.K. Biju, Y.V.S. Murthy, L.R. Cenkeramaddi, N. Kumari, Weather prediction based on the cloud images using machine and deep learning algorithms, in: *Proceedings of the 2025 International Conference on Next Generation Communication & Information Processing (INCIP)*, Bangalore, India, 2025, pp. 937–942, <https://doi.org/10.1109/INCIP64058.2025.11020379>.
- [37] F. Alqahtani, AI-driven improvement of monthly average rainfall forecasting in Mecca using grid search optimization for LSTM networks, *J. Water Clim. Chang.* 15 (4) (2024) 1439–1458, <https://doi.org/10.2166/wcc.2024.242>.
- [38] M.S. Rahman, F.A. Tumpa, M.S. Islam, A. Al Arabi, M.S. Bin Hossain, M.S. Ul Haque, Comparative evaluation of weather forecasting using machine learning models, in: *Proceedings of the 2023 26th International Conference on Computer and Information Technology ICCIT 2023*, 2023, <https://doi.org/10.1109/ICCIT60459.2023.10441077>.
- [39] A. Hussain, A. Aslam, S. Tripura, V. Dhanawat, V. Shinde, Weather forecasting using machine learning techniques: rainfall and temperature analysis, *J. Adv. Inf. Technol.* 15 (12) (2024) 1329–1338, <https://doi.org/10.12720/jait.15.12.1329-1338>.
- [40] Q.X. Jiajiang Shen, Weiyan Wu, Accurate prediction of temperature indicators in Eastern China using a multi-Scale CNN-LSTM-attention model, *Mach. Learn.* (2024), <https://doi.org/10.48550/arXiv.2412.07997>.
- [41] Y.Q. Bangyu Li, Weather prediction using CNN-LSTM for time series analysis: a case study on delhi temperature data, *Mach. Learn.* (2024), <https://doi.org/10.48550/arXiv.2409.09414>.
- [42] H. Du, et al., WeatherMesh-3: fast and accurate operational global weather forecasting, *Mach. Learn.* (2025) [Online]. Available: <http://arxiv.org/abs/2503.22235>.
- [43] S. Pande, B. Banerjee, HyperLoopNet: Hyperspectral image classification using multiscale self-looping convolutional networks, *ISPRS J. Photogramm. Remote Sens.* 183 (2022) 422–438, <https://doi.org/10.1016/j.isprsjprs.2021.11.021>.
- [44] L.Z. Yong, S. Khairunniza-Bejo, M. Jahari, F.M. Muharam, Automatic disease detection of basal stem rot using deep learning and hyperspectral imaging, *Agriculture* 13 (1) (2023), <https://doi.org/10.3390/agriculture13010069>.
- [45] H. Zhang, et al., Hyperspectral image classification on large-scale agricultural crops: the heilongjiang benchmark dataset, validation procedure, and baseline results, *Remote Sens.* 16 (3) (2024), <https://doi.org/10.3390/rs16030478>.
- [46] M.K. Dahouda, I. Joe, Neural architecture search net-based feature extraction with modular neural network for image classification of copper/cobalt raw minerals, *IEEE Access* 10 (2022) 72253–72262, <https://doi.org/10.1109/ACCESS.2022.3187420>.
- [47] A.T. Kumar, et al., Study on the impact of major irrigation projects on rice cultivation in Telangana using remote sensing, *Int. J. Res. Agron.* 7 (10) (2024) 81–91, <https://doi.org/10.33545/2618060x.2024.v7.i10b.1701>.
- [48] M. De Silva, D. Brown, Multispectral plant disease detection with vision transformer-convolutional neural network hybrid approaches, *Sensors (Basel)* 23 (20) (2023), <https://doi.org/10.3390/s23208531>.
- [49] A.K. Singh, A. Rao, P. Chattopadhyay, R. Maurya, L. Singh, Effective plant disease diagnosis using vision transformer trained with leafy-generative adversarial network-generated images, *Expert Syst. Appl.* 254 (2024), <https://doi.org/10.1016/j.eswa.2024.124387>.
- [50] S. Yu, L. Xie, Q. Huang, Inception convolutional vision transformers for plant disease identification, *Internet Things (Netherlands)* 21 (2023), <https://doi.org/10.1016/j.iot.2022.100650>.